

AUTONOMOUS CRACK DETECTION ROBOT FOR INDUSTRIAL INFRASTRUCTURE INSPECTION

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INTRODUCTION

In the previous phase of the project, an autonomous crack detection system was successfully developed using the TurtleBot3 Waffle Pi integrated with ROS 1 Noetic, Gazebo simulation and a YOLOv8 deep learning model (Ibrahimi 2024). The robot was capable of capturing camera streams, performing real-time crack detection and navigating autonomously within a simulated inspection environment. This system demonstrated strong detection accuracy and effective integration between perception, simulation and robotic control (Marian et al. 2020).

However, while the system performed well at detecting cracks, its operational intelligence remained limited. The robot could identify cracks and announce their presence but it lacked the ability to interpret detection results, reason about severity, respond intelligently to inspection-related questions or provide meaningful inspection reports (Lee et al. 2023). These limitations reduced the system's usefulness for real-world inspection and maintenance decision-making.

To overcome these challenges, this project introduces an intelligent reasoning extension through the integration of a Large Language Model (LLM). Specifically, the Google Gemini LLM is incorporated to enhance the robot's autonomy, interpretability and interaction capabilities. By combining computer vision with natural language reasoning, the system is transformed from a basic detection platform into an intelligent inspection assistant aligned with modern AI-driven inspection standards (Cuevas & Manrique 2025).

CURRENT PROBLEM / LIMITATIONS

Although the autonomous crack detection robot can navigate automatically on user command and announcing when a crack is detected, the current system still exhibits several important limitations. Firstly, the robot's response to crack detection is fixed and shallow. When a crack is detected, the system only produces a simple notification such as "Crack detected" without explaining the crack type, severity level or potential structural risk. This limits the usefulness of the detection results for inspection analysis and maintenance planning.

Secondly, the robot does not perform reasoning over detection results. While the YOLOv8 model outputs confidence scores and bounding box information, these values are not interpreted further to determine whether a crack is minor, moderate or severe. The system also does not decide what follow-up action should be taken based on detection quality or risk level.

Thirdly, human-robot interaction is limited. Although the robot can move automatically in response to low-level navigation commands, it cannot respond to higher-level inspection-related queries such as:

- "How serious is the detected crack?"
- "Should the robot continue scanning or stop?"
- "Summarize the inspection results so far."

Finally, the system lacks inspection reporting capability. All outputs remain at the raw detection level, and no structured, human-readable inspection summary or maintenance recommendation is generated. As a result, the robot functions primarily as a detection tool rather than an intelligent inspection assistant. These limitations motivate the need for an intelligent reasoning extension to improve system autonomy, interpretability and usability.

LLM-BASED SOLUTION (GEMINI REASONING MODULE)

To address the identified limitations, a Gemini-based Large Language Model (LLM) reasoning module is integrated (Kim et al. 2024) into the existing Gazebo-based autonomous crack detection system. This module enhances the system by introducing intelligent reasoning and natural language understanding capabilities that were absent in the original implementation. The Gemini LLM functions as an intelligent decision-making layer (Sun et al. 2025) positioned after the YOLOv8 crack detection module. Instead of merely announcing that a crack has been detected, the system now forwards structured detection data such as crack type, confidence score and spatial location to the LLM for deeper analysis and interpretation. This enables the robot to understand the significance of detected cracks rather than treating all detections equally. Using this information, the LLM performs crack interpretation and severity analysis by evaluating detection confidence and observed crack patterns. Detected cracks are classified into severity levels (Ai et al. 2018), namely minor, moderate or severe, providing meaningful context for inspection and maintenance decision-making.

In addition, the system delivers reasoned verbal feedback. When a crack is detected, the robot produces informative explanations rather than a basic alert. For example, it may report that a severe longitudinal crack has been detected and explain the associated structural risk, thereby improving situational awareness for human operators. Based on the assessed severity, the Gemini module also generates action recommendations (Raptis et al. 2025) for the robot. These recommendations may include continuing the inspection, moving closer to the detected area for more detailed scanning or stopping the inspection if safety concerns are identified. This significantly improves the robot's autonomous decision-making capability.

Furthermore, the integration enables natural language command handling. The robot can now understand and respond to higher-level user commands such as requesting an inspection summary, reporting only severe cracks or asking what action should be taken next. This enhances human-robot interaction and makes the system more intuitive to use.

Finally, the Gemini LLM generates structured, human-readable inspection summaries that include detected crack types, severity levels, risk assessments and suggested maintenance actions. Through this integration, the robot evolves from a system that merely moves and announces detections into an intelligent inspection robot capable of reasoning about structural conditions and communicating valuable inspection insights.

OBJECTIVES

The primary objective of this project is to enhance an existing autonomous crack detection system by integrating a Gemini-based Large Language Model (LLM) to introduce intelligent reasoning, improved interaction and structured inspection reporting. Specifically, the objectives of this project are to:

1. **Command Interpretation and Interaction**
Integrate the Gemini LLM to interpret both human-issued natural language commands and perception data derived from YOLOv8 outputs, enabling the robot to correctly respond to at least five high-level inspection commands, including inspection summarization, severe crack reporting, next-action suggestion, crack type listing and risk identification. (Pang et al. 2023).
2. **Intelligent Analysis and Reporting**
Analyze crack detection results and generate structured, human-readable inspection summaries that include crack type, severity level, risk assessment and recommended maintenance actions, with a target relevance accuracy of at least 85% based on manual evaluation.
3. **Real-Time Reasoning Performance**

Ensure that the integration of the Gemini LLM within the Gazebo-based autonomous inspection simulation maintains an LLM reasoning response time of less than few seconds per query without disrupting real-time crack detection or robot navigation.

MODULE DESCRIPTION

The newly added module is a Large Language Model (LLM) reasoning system powered by Google Gemini, developed by Google DeepMind. Gemini is designed for advanced reasoning, multimodal understanding and structured analysis (Team et al. 2023), making it suitable for robotics applications that require interpretation of perception data and natural language interaction.

In this project, the Gemini model is accessed through a cloud-based inference environment such as Google Colab (Team et al. 2023) using the Gemini API. This deployment approach allows efficient GPU-accelerated inference while avoiding the computational constraints of on-board execution. Gemini provides strong natural language understanding, reasoning and summarization capabilities which are essential for translating raw detection outputs into actionable insights (Vavekanand & Sam 2024).

The inputs to the Gemini reasoning module consist primarily of structured YOLOv8 crack detection outputs formatted in JSON. These include crack type (longitudinal, transverse), confidence values representing detection certainty and bounding box information specifying crack location and size. These machine-generated inputs may be supplemented by optional human commands, such as “Explain the severity”, “What should the robot do next?” or “List all detected cracks”, allowing flexible interaction and task specification.

The Gemini module produces several key outputs to support automated pavement inspection. First, it generates an inspection summary that outlines the overall surface condition, including the number, type and severity of detected cracks. This is followed by a structural risk assessment estimating the potential impact of the detected damage on long-term pavement integrity. Based on this analysis, the system recommends appropriate maintenance actions such as sealing, resurfacing or continued monitoring. When detection confidence is low or visual coverage is insufficient, the system also provides rescan suggestions. Additionally, Gemini generates natural language explanations describing why each crack was flagged and how it may affect structural performance.

The reasoning process begins when Gemini receives JSON-formatted detection results from the vision module. These structured inputs are transformed into contextualized natural language descriptions that allow the LLM to interpret the inspection scenario intuitively. A carefully designed prompt then guides Gemini to analyze severity, assess risk and infer suitable actions. The final output includes estimated severity levels, justification for risk classification (minor, moderate or severe), suggested robot actions such as stopping, approaching the crack or continuing the inspection and maintenance recommendations for human operators (Wang 2025).

INTEGRATION WITH EXISTING SYSTEM

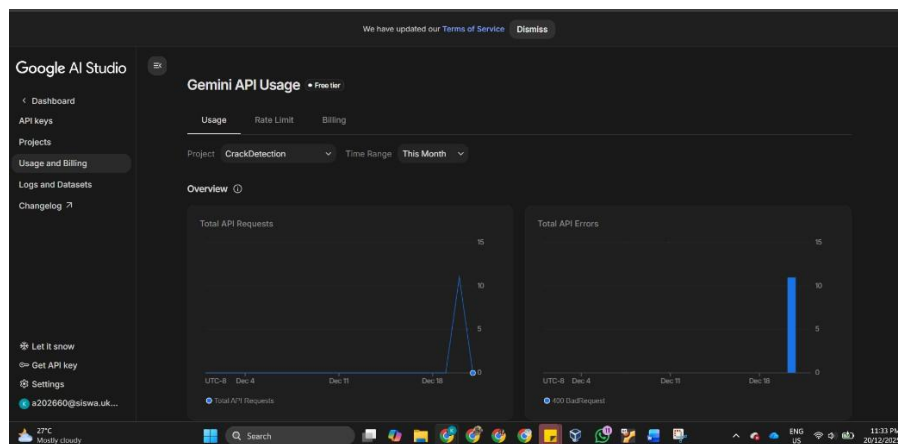


Figure 1

The system's foundation is a YOLO-based crack detection module (`yolo_detector.py`) that processes real-time camera images from the `/camera/rgb/image_raw` topic. The module detects cracks in the images, draws bounding boxes around them, and displays the confidence score for each detection. This allows the user to visually identify where cracks are present and how confident the model is in each detection. The annotated images are published to `/camera/yolo_image`, enabling visualization in tools like `rqt_image_view`.

The Gemini module (`llm_decision_node.py`) has been integrated to provide intelligent analysis on top of YOLO's detections. When a crack is detected, the cropped image of the crack is published to a dedicated topic, `/crack_detection/crack_image`. The Gemini module subscribes to this topic and sends the crack image to the Gemini API using the `google.generativeai` library. The API analyzes the crack and returns a severity classification: MINOR, MODERATE, or SEVERE. This adds a layer of AI reasoning that interprets the visual detection, rather than simply relying on bounding boxes.

The integration ensures that YOLO continues to detect cracks in real-time while Gemini evaluates their severity without delaying the camera feed or affecting robot movement. The severity results are published to `/crack_detection/severity`, allowing other nodes or the operator to access them. The system is verified through API usage logs, which show that calls to the Gemini service are being made each time a crack is detected. This confirms that the integration is working as intended, combining real-time visual detection with AI-based analysis.

DEMONSTRATION PLAN

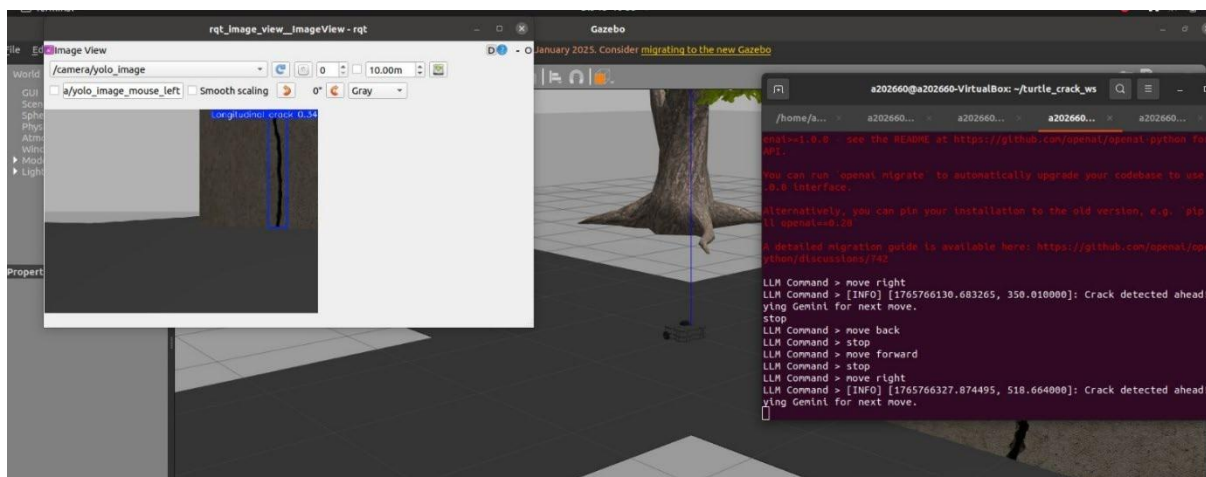


Figure 2

For demonstration, the robot is manually navigated through an environment containing cracks. As it moves, the YOLO detector continuously processes the live camera feed to detect cracks, draw bounding boxes, and display confidence scores. This allows the operator to visually locate cracks and assess detection accuracy in real-time.

When a crack is detected, the cropped crack image is automatically sent to the Gemini API through the `llm_decision_node.py` module. Gemini analyzes the crack and returns its severity, which is published to `/crack_detection/severity`. The operator can immediately see the classification (MINOR, MODERATE, or SEVERE), enabling informed decisions on whether the robot should stop, avoid, or continue inspection in that area. This showcases the real-time AI reasoning integrated with the manual inspection workflow.

The Gemini API significantly enhances the system by adding a cognitive layer to the purely visual YOLO detection. While YOLO identifies cracks and their locations, Gemini interprets the significance of each crack, providing actionable insights without human guesswork. The API usage logs, along with visual feedback from the annotated camera images, demonstrate the system's ability to combine perception and intelligence seamlessly. This integration exemplifies the potential of AI-assisted robotic inspection for real-world applications.

CONCLUSION

The integration of the Gemini Large Language Model represents a significant advancement in the autonomous crack detection system, transforming it from a purely perception-driven robot into an intelligent inspection assistant capable of understanding, analyzing and communicating inspection results. By addressing the system's original limitations in reasoning, interaction and reporting, the Gemini-based extension enables informed decision-making, improved usability and enhanced autonomy.

This successful integration demonstrates the feasibility of incorporating LLM-based reasoning into robotics pipelines (Khan et al.) using ROS and cloud-based inference platforms. It also provides a strong foundation for future work, including autonomous report generation, tighter real-time integration and multimodal reasoning that combines vision, language and motion planning for advanced robotic inspection tasks.

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SIMULATION VIDEO

https://drive.google.com/drive/folders/18_Ry8lzgs1xATEp1EAL7j1pLE87wcBR7?usp=sharing